

Department of Economics
Jahangirnagar University
M.Sc. Final Examination 2021
Course No. Econ 501 (Advanced Microeconomic Theory)
Time : 4 Hours Full Marks : 70

Answer any FIVE (5) questions by taking **at least one** question from **each section**. All parts of a particular question must be answered consecutively.

Section A: General Equilibrium

Question No. 1

(7 + 7 = 14)

- a) Show that
- i) existence of an equilibrium occurs at a positive price
 - ii) stability depends on the relationship between the slopes of the demand and supply curves, and
 - iii) uniqueness is related to the slope of the excess demand function.
- b) 'Even if the individual demand functions are not continuous, existence of large number of agents will guarantee continuity of aggregate excess demand function and therefore existence of equilibrium' - explain the statement.

Question No. 2

(8 + 6 = 14)

- a) Define externality. Graphically explain different types of consumption externalities. Do the externalities create any problem to attain general equilibrium solution in a $2 \times 2 \times 2$ model? Explain.
- b) Explain the three major problems that arise in connection with a general equilibrium.

Question No. 3

(8 + 6 = 14)

- a) Show that how a simple economy with perfectly competitive production and factor markets have an inescapable tendency toward a general equilibrium solution.
- b) Mathematically show and interpret that in perfectly competitive settings:

$$MRPT_{x,y} = \frac{MC_x}{MC_y} = \frac{P_x}{P_y}$$

Section B: Welfare Economics

Question No. 4

(6 + 8 = 14)

- a) Explain Hicks-Kaldor compensation principle of attaining social welfare. How has Scitovsky corrected the contradiction arisen in this principle.
- b) Derive the grand utility possibility frontier and determine the 'point of bliss'. Does such point ensure maximum social welfare? Justify your answer.

Question No. 5

(4 + 4 + 6 = 14)

- a) Define Pigou's ideal level of output. Does this level of output ensure social welfare maximization in the presence of consumption/production externality?

- b) Can Pigou's ideal output level be considered as the first best solution to the maximization of welfare? Examine.
- c) 'Even if individual preference ranking satisfies the property of consistency, it may not give us a consistent ranking of social preference'. Explain in line with the Arrow's impossibility theorem.

Section C: Externality and Asymmetric Information

Question No. 6

(4+5+5=14)

- Differentiate between 'price (tax or subsidy) and quantity (regulation) approaches' to addressing externalities.
- Show that when firms have different marginal costs of pollution reduction, price mechanisms are more efficient means of accomplishing environmental goals unless quantity regulation is accompanied by the ability to meet regulatory targets by trading pollution permits across polluters.
- Evaluate the market for pollution reduction with uncertain costs by considering two extreme cases of externalities: global warming and nuclear leakage.

Question No. 7

(5 + 4 + 5 = 14)

- What are the informational asymmetries in the market models? Examine with examples.
- Why might a labor market unravel? Discuss in brief.
- In Bangladesh, bus drivers have at times been paid based on the number of riders they pick up. Discuss the likely differences in driver behavior under this plan compared to a plan in which drivers are paid by the day.

Question No. 8

(7 + 7 = 14)

- What do you mean by separating equilibrium and pooling equilibrium?

Suppose the value of a worker's marginal product is \$55 per hour for high-ability workers and \$15 per hour for low-ability workers. The preferences of high-ability workers correspond to the utility function $U_H(E, W) = W - 4E$, and the preferences of low-ability workers correspond to the utility function $U_L(E, W) = W - 8E$. By law, everyone is required to attend at least 10 years of school. One-quarter of the population has low ability, and three-quarters have high ability.

What can you say about the educational attainment of each type of worker in a separating equilibrium, and in a pooling equilibrium? Are the workers better off with the most efficient separating equilibrium, or with the most efficient pooling equilibrium?

- What do you mean by incentive scheme?

Suppose a salesperson works for a car dealership for 40 hours per week, but may not choose to work hard all of the time. The dealership's owner cannot observe the salesperson's effort, but can observe the number of cars sold. The salesperson's personal cost of working at the dealership is $C = 1,000 + H^2$, where H is the number of hours during which he works hard. The corresponding marginal cost is $MC = 2H$. Without any effort, the salesperson will, on average, generate a profit of \$800. With each hour of high effort, he has a 10 percent chance of selling a car. Each car sale generates a profit of \$1,000.

What is the efficient number of hours of high effort? How much surplus does the relationship generate?

Answer any five of the following questions. All parts of a particular question must be answered consecutively.

1.

- (a) Construct a simple Neoclassical growth model incorporating Harrod-neutral technological progress, and show that equilibrium values of per capita output y and capital stock k exist. [6]
- (b) Show the impact of saving ratio in the steady state solution and prove that exogenous change in the saving ratio leave only level effects on k and y , but not growth effects. [4]
- (c) Show the impacts of changes in the saving ratio on per capita consumption and long run output. Does any steady state solution automatically imply golden rule of capital accumulation? Discuss. [4]

2. Consider the standard Ramsey model of infinite-horizon households with the following set of preferences.

$$\int_0^{\infty} u(c_t) e^{nt} e^{-\rho t} dt$$

$$u(c) = \frac{c^{1-\sigma} - 1}{1-\sigma}$$

The population growth rate is n , there is the typical neoclassical production function, and technology grows at g rate. The representative household inelastically supplies one unit of labor for real wage w and earns interest r from saving. The representative firm does not own capital and hence solves a static profit maximization problem. Now answer the following questions.

- (a) Set up the Household's problem and find the Euler equation. [4]
- (b) Assume that interest rate r is constant. Derive the consumption function. [4]
- (c) Solve the representative firm's problem. [3]
- (d) Combine parts (a) and (c) to derive the system of two ordinary differential equations (ODEs) and characterize the equilibrium in the model. [3]

3. Consider the Solow economy which accumulates both physical (K) and human capital (H). The economy-wide production function is given by

$$Y = K^{\alpha} H^{\beta} (AL)^{\gamma}$$

, which follows CRS and other neoclassical properties. Technology is fixed. Both forms of capital follow the usual law of motion with a depreciation rate given by δ . Households save a fixed percentage of income for physical and human capital formation, denoted by $s_k > 0$ and $s_h > 0$, respectively. Population grows by n rate. Now, answer the following questions.

- (a) Show that the per capita output growth rate is still zero for this Solow economy. [4]
- (b) Show that this economy can experience positive long-run growth in the presence of increasing returns to scale. [3]
- (c) Now consider the Ramsey economy without technological progress. Suppose the Social Planner maximizes

$$\int_0^{\infty} \frac{c^{1-\sigma} - 1}{1-\sigma} e^{nt} e^{-\rho t} dt$$

subject to the resource constraint:

$$y_t = c_t + i_t$$

, where $y_t = Ak_t$, $A > 0$. Solve the Social Planner's problem and find the condition for LR growth. [4]

- (d) Check the compatibility of the BGP equilibrium with the TVC. [3]

4. Consider the infinitely lived representative household that derives utility from both consumption (c_t) and leisure ($1 - l_t$). The production function of the representative firm is given by

$$f(k_t, l_t, z_t) = z_t k_t^{\gamma} l_t^{1-\gamma}$$

, where z_t is the productivity parameter. Assume that z_t follows AR(1) process i.e., $z_{t+1} = \phi z_t + \epsilon_{t+1}$, where ϵ_t is a white noise error term.

- (a) Construct the Household's problem and show that MRS between consumption and leisure is the real wage.[2]
- (b) Assuming log-utility show that short run labour supply curve is upward sloping but labor supply is independent of wage in the long run.[5]
- (c) Assuming CRRA utility $u(c, h) = \frac{c^{1-\sigma}}{1-\sigma} - \frac{h^{\frac{\sigma-1}{\sigma}}}{1-\sigma}$, find the elasticity of labor supply.[3]
- (d) Mention two important limitations of early business cycle models. How the idea of extensive margin and indivisible labor supply can help overcome these limitations. Show.[4]

5. Consider a finitely-lived representative consumer who inelastically supplies one unit of labor. This consumer-worker receives wage w for per unit of labor supplied and gets interest r from saving. The problem to this consumer is to choose how much to save and consume over her lifetime. The economy is closed without government and the population does not grow. Now answer the following questions.

- (a) Set up the representative consumer's problem, find the Euler equation, and interpret it. [4]
- (b) Find a closed-form solution of consumption and show the pattern of bond holding with constant labor income. [6]
- (c) Show that the model delivers properties of the life-cycle hypothesis when labor income is non-constant.[4]

6. Consider a Ramsey-Cass-Koopmans model with the household's preference given by

$$\sum_{t=0}^{\infty} \beta^t \log c_t$$

The household may spend any fraction of their unit time endowment to work. And the production technology is given by

$$y_t = k_t^\alpha$$

where y_t satisfies usual assumptions. Output can be used either for consumption or investment. Capital is fully depreciated and thus the law of motion of capital is

$$k_{t+1} = i, \delta = 1$$

The initial capital stock is $k_0 > 0$.

- (a) Set up the social planner's problem for this economy, find the Euler equation, and interpret it.[4]
- (c) Use the variable transformation method to convert the Euler equation into a first order difference equation.[3]
- (d) Characterize the steady state and find the optimal policy function $k_{t+1} = g(k_t)$. [3].
- (e) Using the Arrow-Debreu market structure shows that the market problem and the Social planner's problems are equivalent. [4]

7. Suppose that the representative firm produces output given by the production function $y_t = ak_t^\alpha$. This infinitely lived firm owns capital stock and takes investment decisions to maximize lifetime profit. Assume that the price of output, P , is one and capital depreciates at the rate δ .

- (a) Construct the firm's problem. Find and interpret the user cost of capital.[3]
- (b) Suppose that the firm faces a quadratic adjustment cost, given by

$$C(i_t, k_t) = \frac{i_t^2}{2\gamma k_t}$$

while making an investment decision. Derive and interpret Tobin's q and find an implicit investment schedule.[5]

- (c) From the equilibrium conditions in part (b), find a system of two differential equations in k and q . [2]
- (d) Linearize the system you found in part (c) and discuss the stability issues of the system.[4]

8. Consider a representative household that inelastically supplies one unit of labor. This consumer-worker receives wage w for per unit of labor supplied. In addition to consumption and bond holding, this household also holds money, which increases utility because of its liquidity services. Her problem is to choose how much to consume, save, work, and hold money so as to maximize her lifetime utility. The economy is closed without government and the population does not grow. Now answer the following questions.

- (a) Set up the representative household's problem and derive dynamic IS and LM curves.[6]
- (b) Suppose that the central bank conducts monetary policy according to some interest rate rule given by

$$r = \phi y$$

and the ASC is upward sloping. Using these facts summarize the New Keynesian AD-AS model. [4]

- (c) Suppose that the economy is hit by a positive demand shock. Show that this demand shock has real effects in the short run. Also, derive the expression for the inflation rate. Is there any way to stabilize inflation in the model? Feel free to use additional assumptions to simply solve the model analytically.[4]

Department of Economics
Jahangirnagar University
M. Sc. Final Examination 2021
Course: Econometrics (Econ 504)
Time: 4 Hours Full Marks: 70

- Answer any five questions
- All parts of a question must be answered consecutively
- Marks on the right margin

1. [4+3+4+3=14]
- A. Which one is a serious problem, 'under-fitting a model' or 'over-fitting a model'? Why?
 - B. Explain the Ramsey's RESET test with an example.
 - C. How does Errors of Measurement in the Explanatory Variable X create problem in a regression model? Explain.
 - D. When do we use Akaike's Information Criterion (AIC) to select a model?

2. [4+4+6=14]
- A. Explain the Linear Probability Model (LPM). What are the usual problems inherent in the LPM? How can we solve them?
 - B. How do we estimate the Logit model when we have grouped data? Explain with an example.
 - C. From data of 54 standard metropolitan statistical areas, the following Logit model was estimated to explain the high murder rate versus the low murder rate:
 $\ln \hat{O} = 1.1387 + 0.0014 P_i + 0.056 C_i - 0.4050 R_i$
 $Se = (0.0009) \quad (0.0227) \quad (0.1568)$
 Where, O = the odds of a high murder rate, P = 2012 population size in thousands,
 C = population growth rate from 2005 to 2012, R = reading quotient, and Se = standard errors.
 - i. Interpret the estimated Logit model.
 - ii. How does a unit increase in the reading quotient affect the odds of having a higher murder rate?
 - iii. How does a percentage point increase in the population growth rate affect the odds of having a higher murder rate?

3. [4+4+6=14]
- A. In the Panel data estimation, what problems are caused if the unobserved time-invariant individual-specific effects are included in the error term?
 - B. How can the 1st difference approach solve the above problem and what are its limitations?
 - C. How would you estimate panel data regression models when:
 - i. The intercept and slope coefficients are constant across time and space, and the error term captures the differences over time and space.
 - ii. The slope coefficients are constant, but the intercept varies over individuals.
 - iii. The slope coefficients are constant, but the intercept varies over individuals and time.

4. [3+3+4+4=14]
- A. What is the role of lags in economics? Write and explain two economic models as examples.
 - B. Distinguish between distributed-lag model and autoregressive lag model. Give examples.
 - C. How did Koyck propose an ingenious method to estimate distributed-lag models?
 - D. Granger (1969) proposed to test the 'causality' between variables. Explain the procedure using an appropriate economic example.

5. [3+5+6=14]
- A. What is a simultaneous equation system? Explain it with an example.
 - B. What problem(s) may arise if we apply the OLS method in a simultaneous equation system? Explain and show your necessary derivation.
 - C. Verify the following equation whether they are identified by both identification rules or not?

Equation no.	1	Y_1	Y_2	Y_3	Y_4	X_1	X_2	X_3
1	$-\beta_{10}$	1	$-\beta_{12}$	$-\beta_{13}$	0	$-\gamma_{11}$	0	0
2	$-\beta_{20}$	0	1	$-\beta_{23}$	0	$-\gamma_{21}$	$-\gamma_{22}$	0
3	$-\beta_{30}$	$-\beta_{31}$	0	1	0	$-\gamma_{31}$	$-\gamma_{32}$	0
4	$-\beta_{40}$	$-\beta_{41}$	$-\beta_{42}$	0	1	0	0	$-\gamma_{43}$

6.

[8+6=14]

A. What is meant by the ILS method? Given the demand-supply model:

$$\text{Demand function: } Q_t = \alpha_0 + \alpha_1 P_t + \alpha_2 X_t + u_{1t}$$

$$\text{Supply function: } Q_t = \beta_0 + \beta_1 P_t + u_{2t}$$

Mention the different steps of estimation using the ILS method. What properties does the ILS estimation hold?

B. What do you mean by 'instrumental variable'? Using a simultaneous model, show the estimation procedure by following the 2SLS method.

7.

[2+5+5+2=14]

A. What do you mean by stochastic process and stationary process?

B. What is Cointegration? Discuss the Engle-Granger two-step method for testing Cointegration.

C. What is a Vector Autoregression (VAR) model? Explain with an example mentioning its limitations.

D. Given the Dickey-Fuller (DF) test results of a variable on time series data below, comment on the results.

Dickey-Fuller test for unit root

	Test Statistic	Interpolated Dickey-Fuller ---		
		1% Critical Value	5% Critical Value	10% Critical Value
Z(t)	-1.368	-3.696	-2.978	-2.620

MacKinnon approximate p-value for Z(t) = 0.6349

8.

[5+3+4+2=14]

A. What is 'unit root' problem in a time series data? Explain the steps of Augmented Dickey-Fuller (ADF) for unit root problem.

B. What is an error correction mechanism (ECM)? What is its relationship with cointegration?

C. From the BJ methodology, explain the identification step to find the value of p, d, q for the model using correlogram and partial correlogram.

D. What are the criteria for lag selection? Given the results of a variable on time series data, how many lags should be chosen and why?

lag	LL	LR	df	p	FPE	AIC	HQIC	SBIC
0	-21.7346				.000065	1.71564	1.7754	1.90246
1	126.437	296.34	16	0.000	9.9e-09	-7.09581	-6.79697	-6.16168
2	148.47	44.066	16	0.000	7.0e-09	-7.49799	-6.96009	-5.81656*
3	168.722	40.505*	16	0.001	6.3e-09*	-7.78148*	-7.0045*	-5.35274
4	193.517	49.59	16	0.000	5.0e-09	-8.36781	-7.35176	-5.19176

☺ THE END ☺

Department of Economics
Jahangirnagar University
M.Sc. Final Examination 2021
Course Econ 507: Health Economics

Time: 4 hours

Full Marks: 70

Instructions

- Answer any 05 (Five) questions
- Marks on the right margin

1. [3+4+(3+4)=14]
- a. Define health economics.
 - b. Explain the distinguishing features of health economics.
 - c. Why does the health care market fail? What is the role of the government to rectify these failures? Explain.
2. [(3+4)+7=14]
- a. How does the cost of health care services become a major challenge in the health care sector? Analyse reform proposals to contain costs.
 - b. Critically explain and express your opinion on what choices we must make in health and health care related services.
3. [3+4+4+3=14]
- a. What are the advantages of applying a health production function?
 - b. Explain diminishing returns in a health production function with the following policy objective: decrease infant mortality rate. 'The use of average measures instead of marginal measures, instead of marginal measures can result in an incorrect allocation of resources among health programmes. Why?
 - c. Construct empirical studies of health production functions based on aggregate data and explain their findings.
 - d. Briefly discuss the historical role of medicine and health care in a rising population and decline in mortality rate.
4. [6+(4+4)=14]
- a. In a perfect health insurance market, a risk averse individual will always be willing to pay full and fair premium for health insurance. Explain.
 - b. Show that in the case of adverse selection:
 - I) Pooling equilibrium is not sustainable.
 - II) Separating equilibrium is possible only if the proportion of high-risk individuals is much lower than low-risk individuals in the pool.
5. [3+4+7=14]
- a. Demonstrate a Martin-Feldstein analysis to calculate the welfare losses of excess health insurance.
 - b. Examine the balance between moral hazard and risk protection using employer-sponsored insurance policies.
 - c. Graphically illustrate income transfer effects of insurance. Show the net welfare gain of price distortion under income transfer effects.

6.

[2+(4+4+4)=14]

- a. What is clinical decision analysis?
- b. Consider a patient who is 56 years old with diabetes and heart block. He may have Covid exposure and a 14 day's history of travelling internationally. In his physical examination, physicians assigned a 70% chance Covid positive (prior probability). Sensitivity and specificity of the Covid test are 80% and 90% respectively.
 - I) Calculate posterior probabilities using the Bayes Theorem.
 - II) Perform steps to reach a clinical decision on performing a Covid test. Consider the following utility values for best to worst outcome: 1.00, 0.75, 0.60, and 0.30.
 - III) Graphically present a sensitivity analysis based on two pre-test likelihoods.

7.

[2+(2+4)+(2+2+2)=14]

- a. Describe the importance of using different approaches to measure quantitative and qualitative health status.
- b. Explain the theoretical foundation for QALY measurement. What are the QALY methods for assessing preferences of respondents? Discuss.
- c. What is EQ-5D? Mention use cases to utilise EQ-5D. Explain constructive methods of EQ-5D-3L, ED-5D-5L, and EQ-5D-Y.

8.

[14]

Explain the five-phase implementation of DHIS2 to transform the health information system in Bangladesh.

☺ GOOD LUCK ☺

Department of Economics
Jahangirnagar University
Masters Final Examination - 2021
Course no: ECON 510 (Labor Economics)

Time: 4 hours

Total: 70

Answer any of five (5) questions from the following, taking at least two (2) from each of part A & part B.
All parts of a question must be answered consecutively.

Part A

- (5+6+3)
1. (a) "Time and monetary costs associated with working affect the shape of budget constraint thus influence the number of hours a person chooses to work"-Explain using diagram (s).
(b) Why does the male participation rate decline and female participation rate increase in the labor market over the periods? Explain.
(c) How does the cross-substitution effect change in hours of work and hours of leisure for the married couple? Explain.
- (2+6+6)
2. (a) Illustrate the investment decision rules for taking higher education.
(b) Differentiate between general on-the-job training and specific on-the-training. Graphically analyze the benefits and costs of these two training.
(c) Discuss using diagram(s) the signaling failure and the optimal signal when education is used as a screening device.
- (8+6)
3. (a) How do you explain the added worker effect and the discourage worker effect in the labor force growth during recessions? Use graph(s) if necessary.
(b) What are the positive and negative aspects of wage subsidy program? Briefly explain.
- (7+7)
4. (a) Why did the labor market not give rise to the predicted wage?
(b) Briefly explain the various issues in the employee selection.

Part B

- (14)
5. What is assimilation? How does it convergence of the labor market outcomes? According to Lee et. Al (2022), discusses the gender aspect of migrants' assimilation in the Europe.
- (14)
6. What are the basic three components of the General Health Questionnaire (GHQ) index? Briefly explain Belloni et. Al. (2022) findings and the empirical frameworks of the impact of working conditions on mental health for the UK.
- (14)
7. What is the Matthew effect? According to Heckman and Landerso (2022) , identify the lessons for Americans from Denmark about inequality and social mobility.
- (14)
8. "Childcare causes in increase in the labor supply of cohabiting mothers who move towards full-time employment. Despite this, average taxes paid on the extra income is low, lending little support to the argument that parts of the cost of childcare is offset by increases in taxes." Justify these research findings in the light of Andersen & Havnes (2019).